Gravitational Field Strength

Exam code: 0625 & 0972

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Updated on 4 July 2025

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Gravitational field strength of a planet

- The strength of a gravitational field around a planet depends on
  - the mass of the planet
  - the distance from the planet

Gravitational field strength and mass

- The relationship between gravitational field strength and mass is:

The greater the mass of a planet, the greater the strength of the gravitational field at its surface

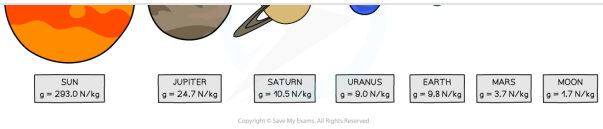
- The value of g (gravitational field strength) varies from planet to planet depending on their mass and radius

Gravitational field strength of bodies in the Solar System

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The strength of the gravitational field at the surface of a planet depends on its mass and radius

Gravitational field strength and distance

- The relationship between gravitational field strength and distance is:

As the distance from a planet increases, the strength of the gravitational field decreases

- At the surface of a planet, the value of g (gravitational field strength) is constant, but it decreases with distance from the planet

Examiner Tips and Tricks

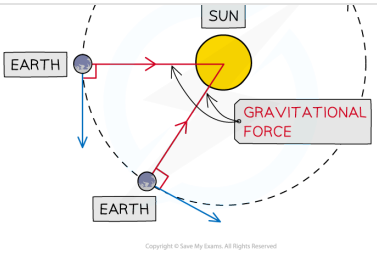
You do not need to remember the value of g on different planets for your exam, and the value of g for Earth is given on the front page of the exam paper.

Gravitational attraction of the Sun

- Orbital motion is a result of the **gravitational force of attraction** acting between two bodies
- This gravitational force
  - always acts **towards the centre** of the larger body
  - causes the orbiting body to move in a **circular path**

- The Sun contains most of the mass (>99%) of the Solar System
- Therefore, for objects orbiting around the Sun
  - the Sun's **gravitational attraction** keeps them in orbit
  - the force is directed from the orbiting object to the centre of the Sun

Orbital motion of the Earth around the Sun



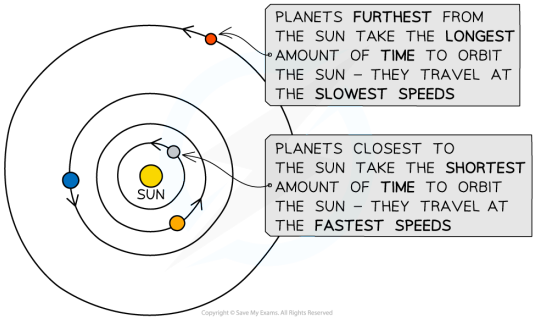
The Sun's gravitational force of attraction keeps the Earth in orbit around the Sun

Orbital motion of planets

Extended tier only

- As the distance from the Sun increases:
  - the Sun's gravitational field strength **decreases**
  - the orbital speed of a planet **decreases**
- For an object to maintain a circular orbit, it must have a **centripetal force**
  - For planets orbiting the Sun, this force is the Sun's **gravitational attraction**
- Therefore, the centripetal force on a planet depends on
  - the strength of the Sun's gravitational field
  - the distance of the planet from the Sun
- The further away a planet is from the Sun, the **weaker** the strength of the Sun's gravitational attraction and the weaker the centripetal force
- The centripetal force on a planet is **proportional** to its orbital speed
- Therefore, the **further** a planet is from the Sun:
  - the **smaller** its orbital speed
  - the **longer** its orbital period

Orbital speed and distance



The closest planets to the Sun have the fastest orbital speeds, and the furthest have the slowest

- This trend in orbital speed and distance can be seen in the data of the planets in the Solar System:

	(million km)	speed (km/s)	(days or years)
Mercury	57.9	47.9	88 days
Venus	108.2	35.0	225 days
Earth	149.6	29.8	365 days
Mars	227.9	24.1	687 days
Jupiter	778.6	13.1	11.9 years
Saturn	1433.5	9.7	29.5 years
Uranus	2872.5	6.8	75 years
Neptune	4495.1	5.4	165 years

Examiner Tips and Tricks

Be careful with your wording in this topic when talking about gravity. It is important to refer to the **force** of gravity as 'gravitational attraction', ' strength of the Sun's gravitational field' or 'the force **due to** gravity'. Avoid terms such as 'the Sun's gravity' or even more vague, 'the force from the Sun'.

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